1845

John Franklin's doomed expedition to find the elusive Northwest Passage through the Arctic Ocean. His two ships HMS *Terror* and HMS *Erebus* vanished after entering Baffin Bay.

ONE OF THE KEY SEARCHES IN SCIENCE TODAY is to discover the underlying unity of forces and matter. In 1845, British physicist **Michael Faraday** made an early contribution to this with a series of experiments in which he showed that a magnetic field could alter the polarization of light as it traveled through heavy lead glass. This finding revealed a previously unsuspected **link between light, magnetism, and electricity**, paving the way for the discovery of the complete spectrum of electromagnetic radiation (see pp.234–35).

At the same time, astronomers were continuing to scrutinize the night sky. An English astronomer, the **3rd Earl of Rosse**, constructed a huge reflecting telescope in Ireland dubbed the Leviathan of Parsonstown. Through its 6 ft (1.8 m) aperture, Rosse observed that the galaxy M51—now known as the Whirlpool Galaxy—had a spiral structure. This was the **first galaxy observed to be spiral**.

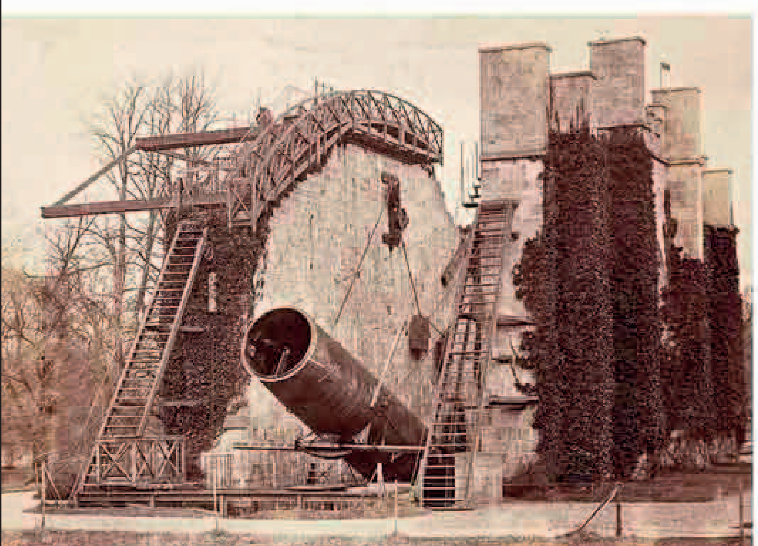
340 TONS

THE WEIGHT OF THE THREE-STORY-HIGH STEAM ENGINE IN THE SS *GREAT BRITAIN*

Other astronomers questioned why Uranus kept appearing in places that it should not appear according to Kepler's laws (pp.100–101) and Newton's laws (pp.120–21). British mathematician **John Couch Adams** (1819–1892) suggested that there was another planet beyond Uranus disturbing its orbit. French astronomer **Urbain Le Verrier** (1811–77) observed disturbances in Uranus's orbit to predict where this planet might be.

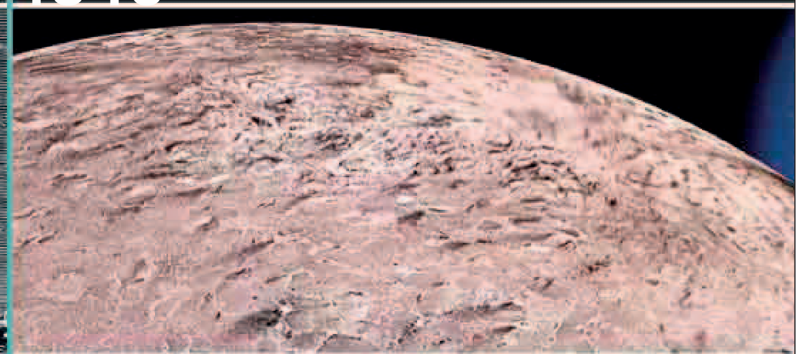
Another mystery was the disappearance of **British explorer John Franklin** and his ships HMS *Erebus* and HMS *Terror*, in Baffin Bay during an expedition to find the Northwest Passage in the Arctic.

The SS *Great Britain* successfully sailed from Liverpool to New York, becoming the first iron steamship driven by a screw-propeller rather than paddle wheels to make this journey across the Atlantic.



Lord Rosse's reflecting telescope *The Leviathan of Parsonstown, constructed in County Offaly, Ireland in 1845, was the biggest telescope in the world for more than 70 years.*

1846



The planet Neptune rising over its moon Triton in an image constructed from pictures taken by the Voyager space probes. Both were discovered in 1846.

GENERAL ANESTHESIA

The introduction of general anesthetics (substances used to render a person unconscious) transformed surgery. It allowed surgeons to perform all kinds of operations painlessly. The term anesthesia means "loss of sensation": anesthetics work by blocking signals that pass along nerves to the brain. The earliest anesthetics were ether, laughing gas (nitrous oxide), and chloroform.

ON SEPTEMBER 23, 1846, German astronomer **Johann Gottfried Galle** (1812–1910) received a letter from French astronomer **Urbain Le Verrier**. It contained instructions telling him where to look to find the solar system's eighth planet, soon to be called **Neptune**. A long dispute followed over who deserved the credit for Neptune's discovery—Urbain Le Verrier or **John Couch Adams**, who had also predicted its existence the previous year. Most authorities credit Le Verrier because he was the one who calculated its position accurately enough for Galle to find it immediately. Within 17 days, English astronomer **William Lassell** (1799–1880) found that Neptune had a moon. It was named Triton a century later.

> " THIS SIGNIFIES **INSENSIBILITY…** TO **OBJECTS OF TOUCH.** THE ADJECTIVE WILL BE **ANESTHETIC.** THUS WE MIGHT SAY, THE **'STATE OF ANESTHESIA'.** "

Oliver Wendell Holmes, US physician, in a letter to Dr. William Morton, 1846

- English astronomer Lord Rosse builds the Leviathan of Parsonstown **telescope**
- English physicist Michael Faraday confirms that **light is electromagnetic**
- British mathematician John Couch Adams and French astronomer Urbain Le Verrier **predict the existence of Neptune**
- **August 10** British engineer Isambard Kingdom Brunel's steamship SS *Great Britain* crosses the Atlantic
- German geographer Alexander von Humboldt publishes **Kosmos**, in which he provides a comprehensive account of the Universe
- Lord Rosse finds that the **nebula M51** has a spiral structure
- British explorer John Franklin's expedition to find the Northwest Passage is lost
- Canadian geologist Abraham Gesnar develops **kerosene (paraffin)** to fuel lamps
- **August 31** French astronomer **Urbain Le Verrier** publishes his predicted orbit for the as yet undiscovered Neptune
- **September 23** German astronomer Johan Galle discovers **Neptune**
- **October 10** English astronomer **William Lassell** discovers Neptune's moon, later named **Triton**